Coupling a 3D Finite-Volume Left Ventricular Model with 0D Lumped Parameter Circulation using OpenFOAM

David Kelly

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European Research Council
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XENOsim

David Kelly
Chanikya Valeti
Philip Cardiff

School of Mechanical and Materials Engineering
University College Dublin

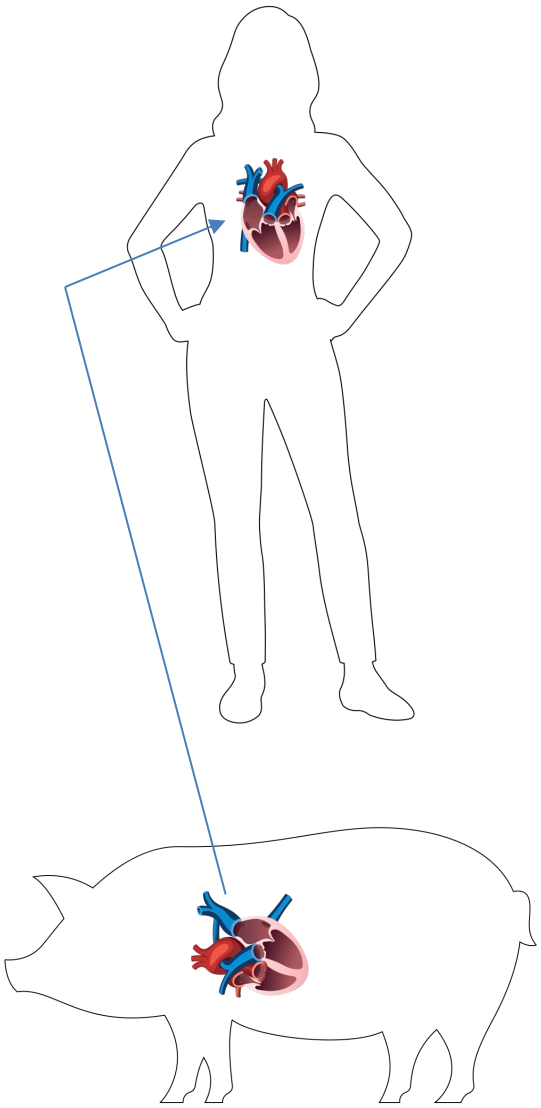
Background & Motivation

Big Picture

- Donor heart shortage continues to limit cardiac transplantation.
- Xenotransplantation is a promising alternative, but haemodynamic performance remains poorly understood.
- Porcine-to-human implantation introduces distinct haemodynamic and electromechanical challenges.

Key Drivers

- Fundamental pig–human differences in anatomy, compliance, and vascular impedance.
- Limited in-vivo haemodynamic access motivates high-fidelity *in-silico* modelling.
- Need for predictive tools to assess pig-heart performance under human circulatory loading.



Completely Lumped System

Lumped-Parameter Element Relations

Resistance (Poiseuille): $\Delta P = RQ$

Compliance: $Q = C\,\dfrac{d(\Delta P)}{dt}$

Inertance: $\Delta P = L\,\dfrac{dQ}{dt}$

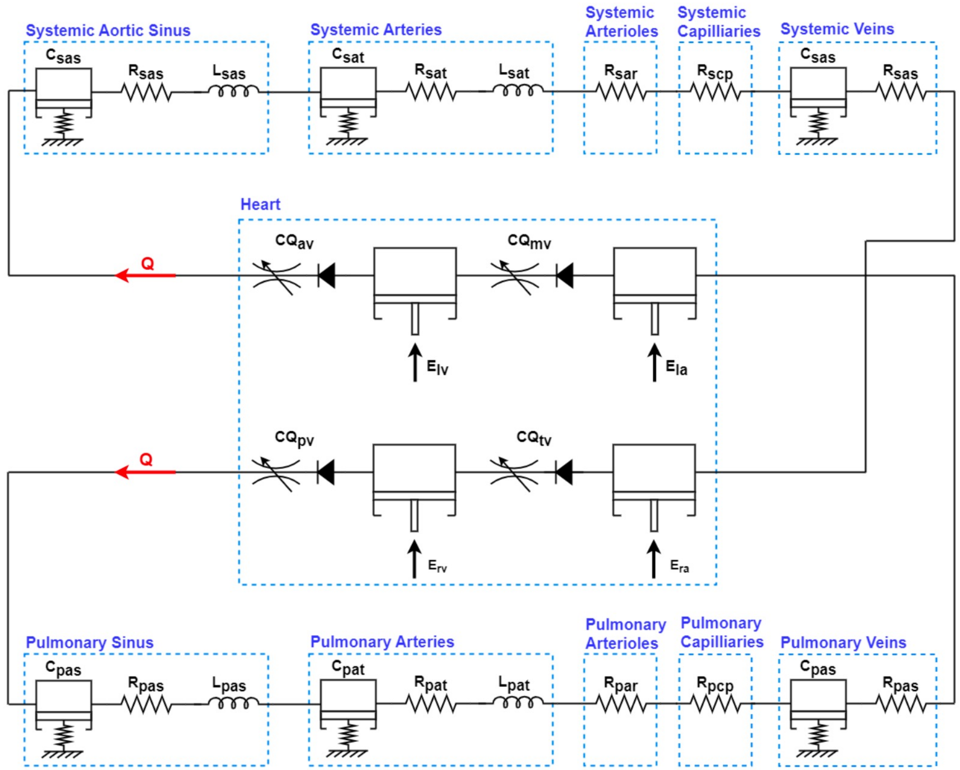
Kirchhoff Laws (Haemodynamic Form)

Flow conservation at a node (KCL):

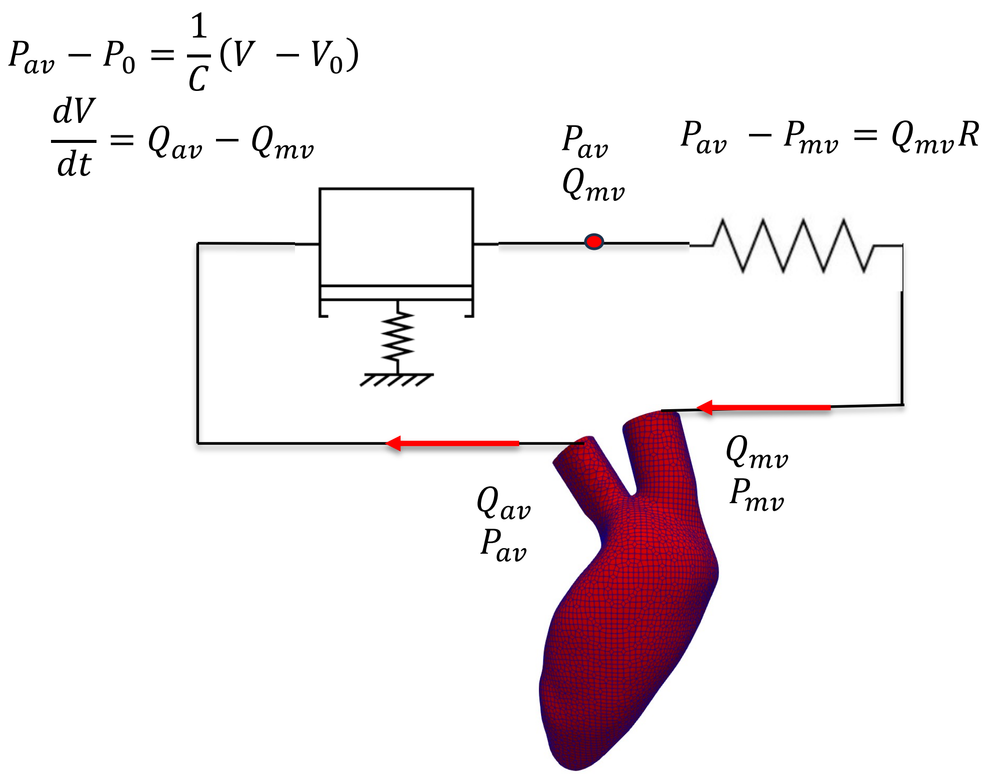
$$\sum Q = 0$$

Pressure balance around a loop (KVL):

$$\sum \Delta P = 0$$



Coupling 3D Ventricle to 0D Circulation



OpenFOAM BCs

- Prescribe P_{av} and Q_{mv} at the OpenFOAM boundaries
- Compute Q_{av} and P_{mv} from the 3D model and use them to update the LPM
- $\frac{dP_{av}}{dt} = \frac{1}{C}(Q_{av} - Q_{mv})$ (setting $P_0 = V_0 = 0$)
- $\frac{dP_{av}}{dt} = \frac{1}{C}(Q_{av} - \frac{1}{R}(P_{av} - P_{mv}))$
- $Q_{mv} = \frac{1}{R}(P_{av} - P_{mv})$

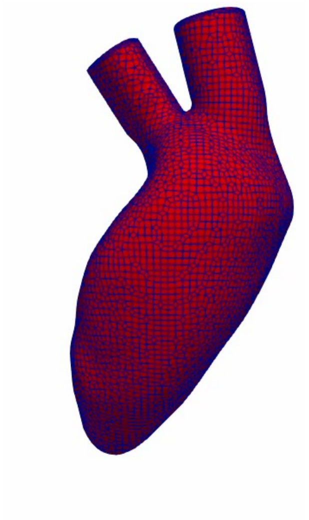
Discretisation and Verifying Closed Loop

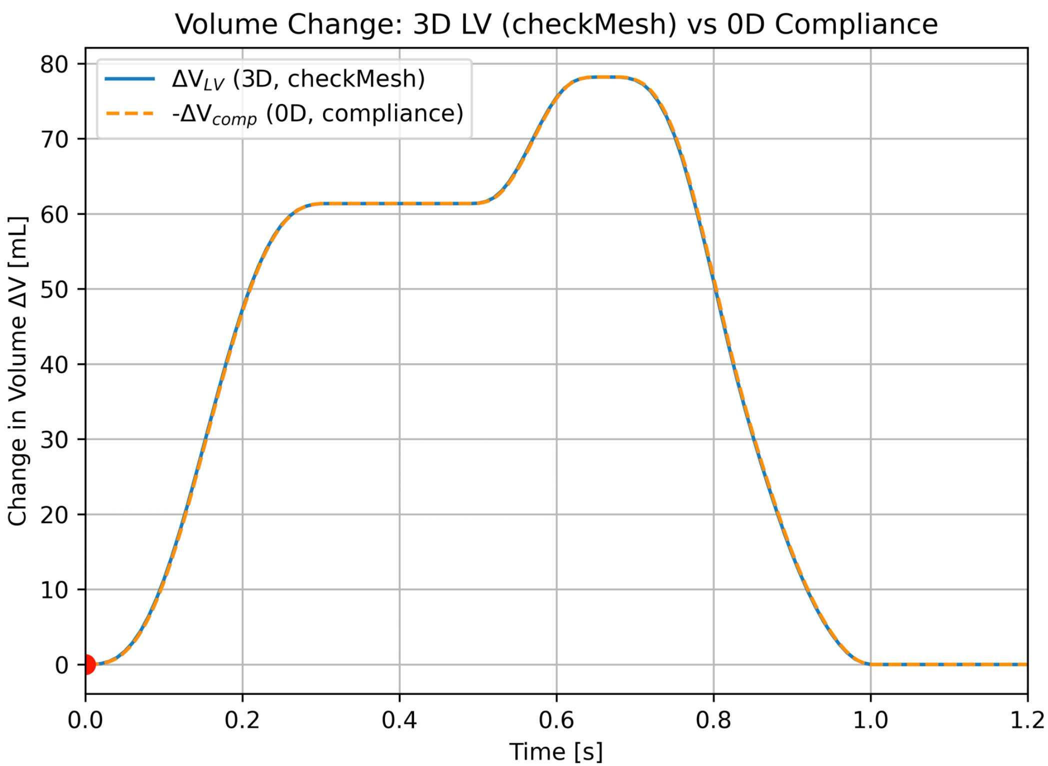
Discretisation

$$P_{av}^{[1]} = \frac{P_{av}^{[0]} + \frac{\Delta t}{C}\left(Q_{av}^{[1]} + \frac{P_{mv}^{[1]}}{R}\right)}{1 + \frac{\Delta t}{RC}}, \quad \text{initial } n = 0$$

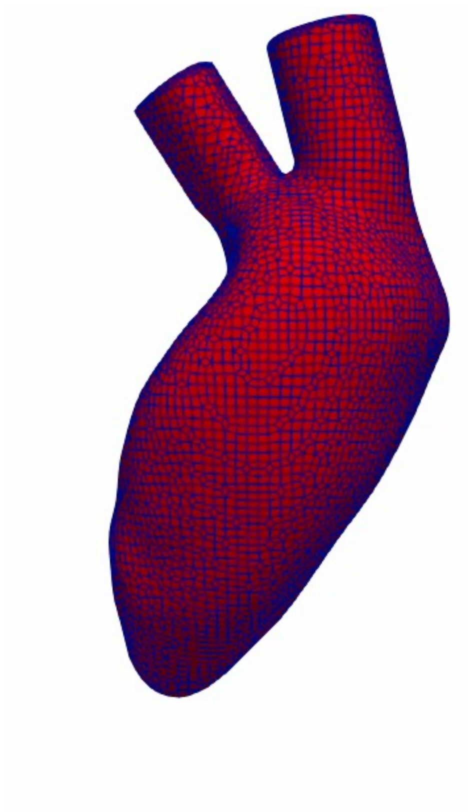
$$P_{av}^{[n]} = \frac{4\,P_{av}^{[n-1]} - P_{av}^{[n-2]} + \frac{2\Delta t}{C}\left(Q_{av}^{[n]} + \frac{P_{mv}^{[n]}}{R}\right)}{3 + \frac{2\Delta t}{RC}}$$

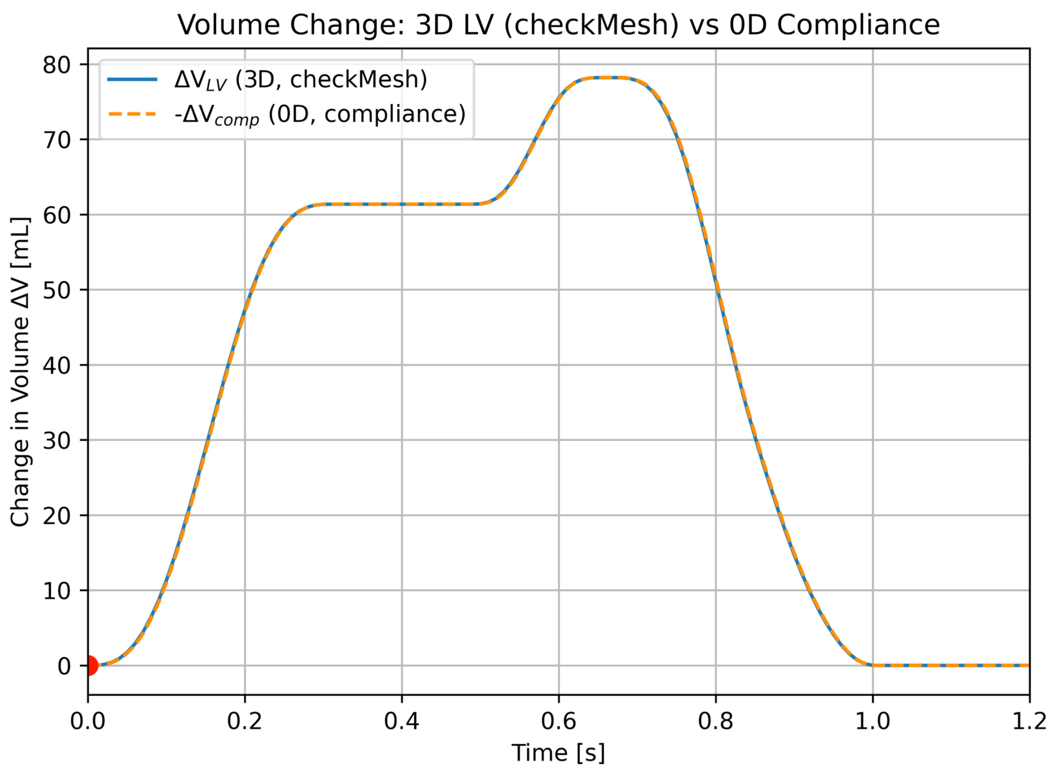
$$Q_{mv}^{[n]} = \frac{P_{av}^{[n]} - P_{mv}^{[n]}}{R}$$



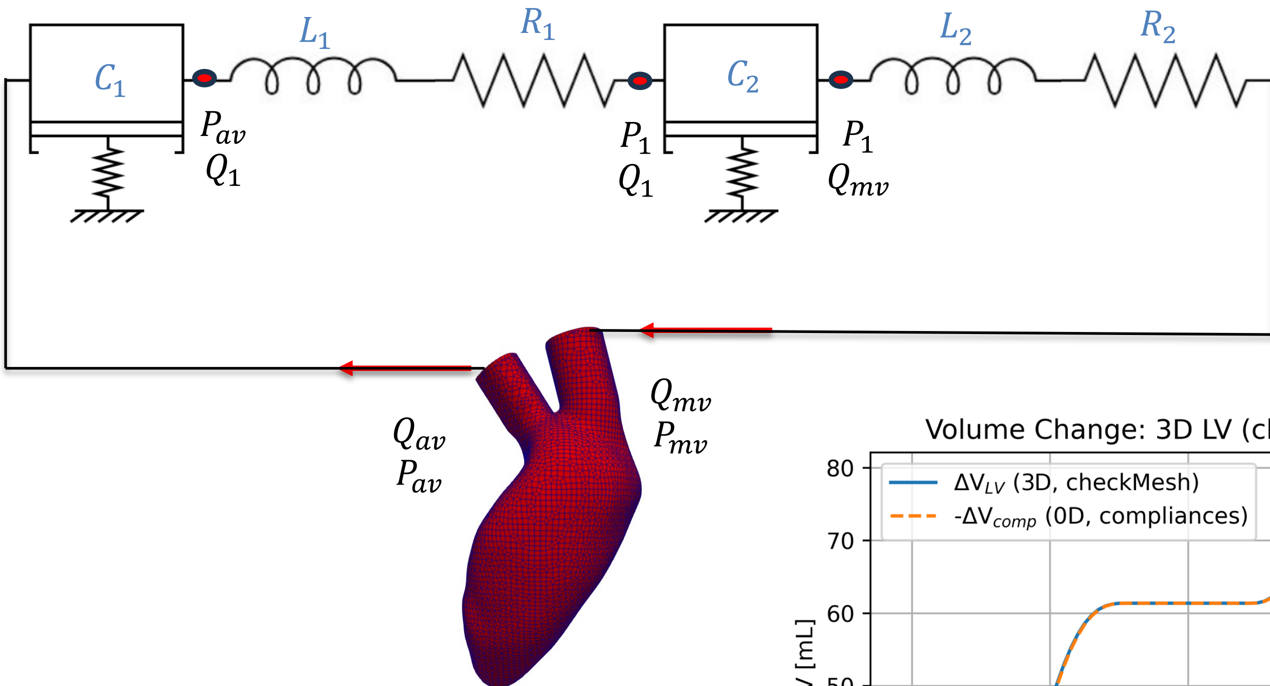


3D Moving Left Ventricle





Coupling to a CLR System

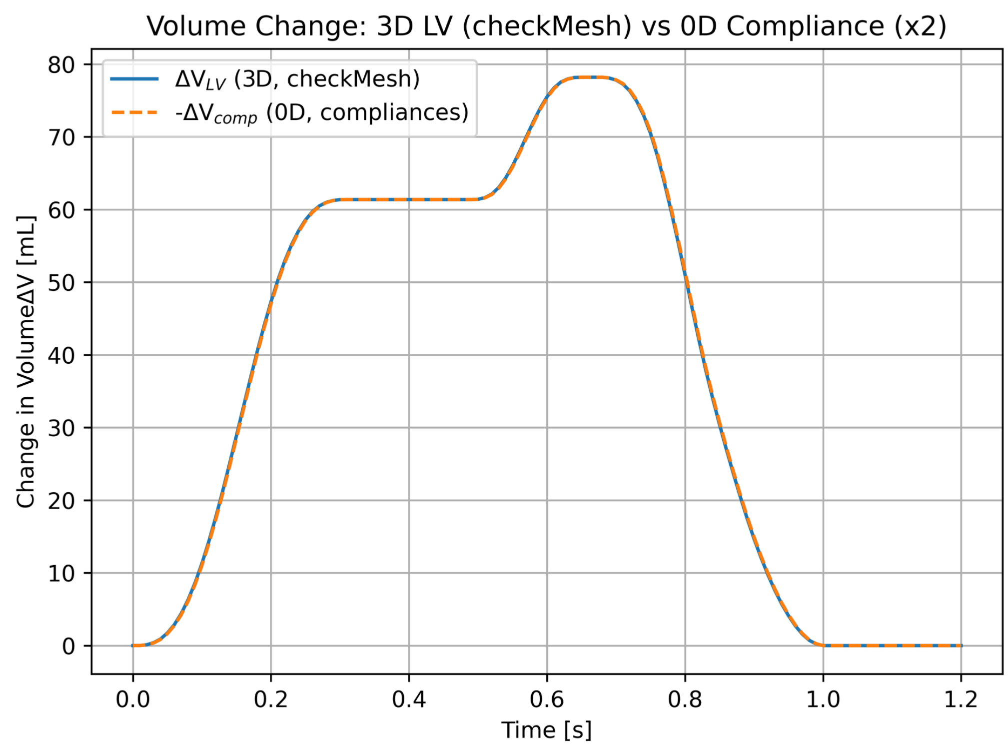


$$\frac{dP_{av}}{dt} = \frac{1}{C_1}\left(Q_{av} - Q_1\right)$$

$$\frac{dQ_1}{dt} = \frac{1}{L_1}\left(P_{av} - P_1 - Q_1 R_1\right)$$

$$\frac{dP_1}{dt} = \frac{1}{C_2}\left(Q_1 - Q_{mv}\right)$$

$$\frac{dQ_{mv}}{dt} = \frac{1}{L_2}\left(P_1 - P_{mv} - Q_{mv} R_2\right)$$



Adding in 0D Chamber and Valve

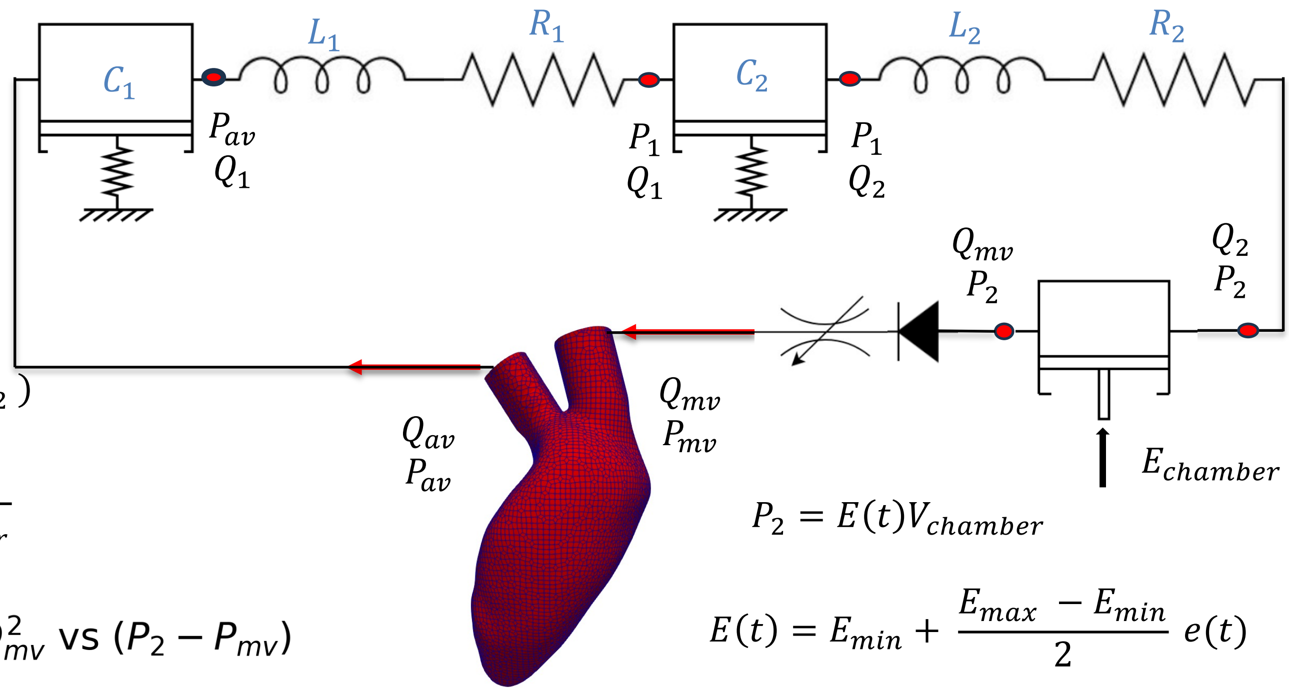
$$\frac{dP_{av}}{dt} = \frac{1}{C_1}\left(Q_{av} - Q_1\right)$$

$$\frac{dQ_1}{dt} = \frac{1}{L_1}\left(P_{av} - P_1 - Q_1 R_1\right)$$

$$\frac{dP_1}{dt} = \frac{1}{C_2}\left(Q_1 - Q_2\right)$$

$$\frac{dQ_2}{dt} = \frac{1}{L_2}\left(P_1 - E(t)V_{chamber} - Q_2 R_2\right)$$

$$\frac{dV_{chamber}}{dt} = Q_2 - CQ\sqrt{E(t)V_{chamber}}$$

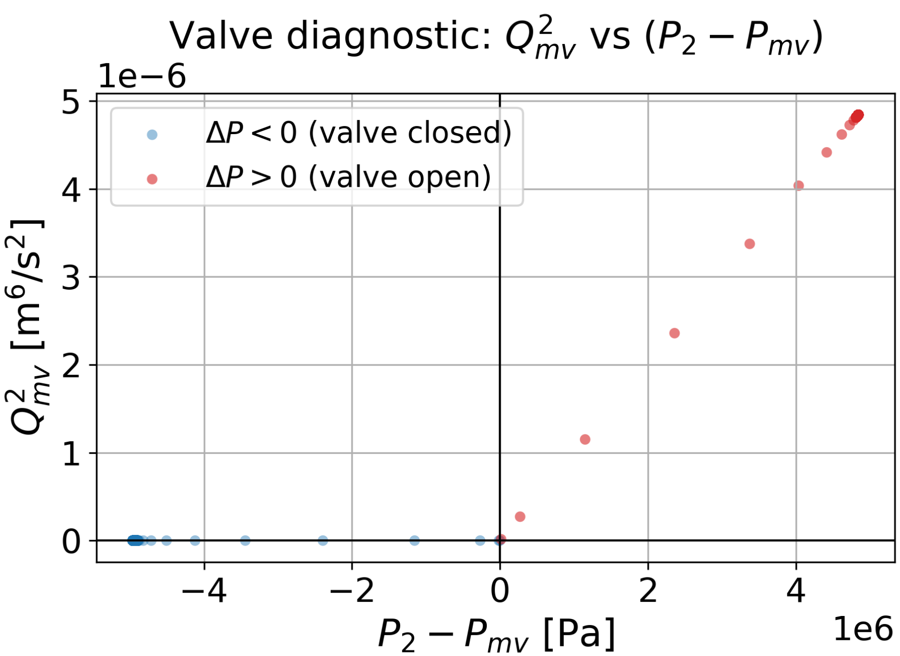


$$P_2 = E(t)V_{chamber}$$

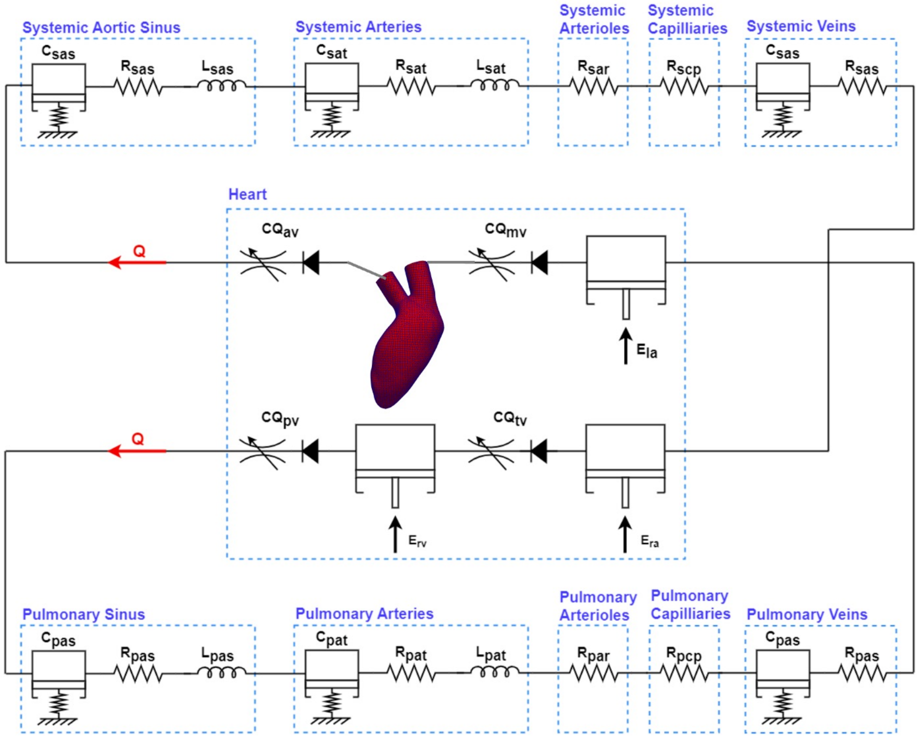
$$E(t) = E_{min} + \frac{E_{max} - E_{min}}{2}\, e(t)$$

$$e(t) = \begin{cases} 1 - \cos\left(\dfrac{t}{T_{s1}}\right) & 0 \le t < T_{s1} \\ 1 - \cos\left(\dfrac{t - T_{s1}}{T_{s2} - T_{s1}}\right) & T_{s1} \le t < T_{s2} \\ 0 & T_{s2} \le t < T \end{cases}$$

$$Q_{mv} = \begin{cases} CQ\sqrt{P_2 - P_{mv}} & P_2 > P_{mv} \\ 0 & P_2 \le P_{mv} \end{cases}$$



Limitations and Future Work



Progress & Next Steps

- Build a full **0D closed-loop circulation** model.
- Add further **3D heart chambers**.
- Incorporate **pig-specific geometry**.
- Develop a **modular OpenFOAM framework** to switch chambers between 0D and 3D.

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David Kelly

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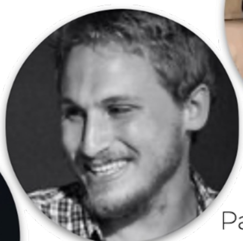
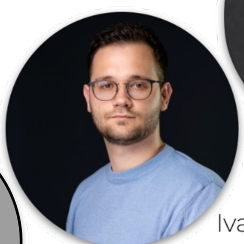


Pablo Castrillo



Ivan Batistić

Chanikya Valeti

Philip Cardiff

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Thank you!

Happy to take questions